Berea Technical College

Department of Engineering Studies
Mathematics N4 Assignment 1

Examiner: Mr Y. Mia **Moderator:** Mr. T. Poovalingam
Total: 50 **Due Date: 30/03/26**

INSTRUCTIONS

1. **Answer ALL QUESTIONS in full. Show ALL the calculations and intermediary steps. Simplify where possible.**
2. **ALL final answers must be accurately approximated to THREE decimal places.**
3. **Questions may be answered in any order.**
4. **All Formulae used must be written down.**

QUESTION 1

1.1 Given:

$$\frac{2p}{3} - 25q = -5$$

$$2p + \frac{5q}{7} - 4 = 0$$

Use Crammer's rule to solve for p and q. **(6)**

1.2 Given:

$$\begin{vmatrix} \frac{1}{2} & -3 & 4 \\ \frac{5}{7} & 1 & -5 \\ \frac{7}{8} & 0 & 6 \end{vmatrix}$$

1.2.1 Determine the minor of 6 **(2)**
1.2.2 Determine the cofactor of -5 **(2)**

1.3 Given:

$$3x + 2y = -5z - 6$$
$$5x + 7y - 2z = 4$$
$$2x - 7z = 5y + 14$$

Solve for x only using Crammer's rule. Expand along ROW1 only. **(10)**

QUESTION 2

2.1 Given: $4x^2 - 12x + 9 = -14$

Solve for x: **(4)**

2.2 Given: $z = -2 - j\sqrt{3}$

2.2.1 **Find $\bar{z}$.** **(1)**

2.2.2 Convert $\bar{z}$ into polar form. θ may only be positive. **(2)**

2.2.3 Indicate $\bar{z}$ and all calculated values on the Argand diagram. **(3)**

2.3 Given: $z = (-4 + 3j)^8$

Simplify by using De Moivre's theorem. Leave the final answer in rectangular form. **(6)**

2.4 Simplify: $-2j^{16} + 3j^9$ **(2)**

2.5 Simplify to rectangular form. Calculations in polar form are excluded

$$\frac{-1 + j7}{-3 + j8}$$ **(4)**

2.6 **Solve for x and y if:**

2.6.1 $$\frac{x - jy}{-1 - j} = \frac{2 - j}{3 + j}$$ **(4)**

2.6.2 $(x - y) + j(x + y) = (1 - j)^2 + j(2 + j)$ **(4)**

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